Proof. (i) Generalized eigenspaces of θ are subrepresentations of W, and W is their direct sum. Thus, θ can have only one eigenvalue λ. If λ is zero, θ is nilpotent; otherwise it is an isomorphism.

(ii) The proof is by induction in n. The base is clear. To make the induction step ($n-1$ to n), assume that θ is not nilpotent. Then by (i), θ is an isomorphism, so $\sum_{i=1}^n \theta^{-1}\theta_i = 1$. The morphisms $\theta^{-1}\theta_i$ are not isomorphisms, so they are nilpotent. Thus $1-\theta^{-1}\theta_n = \theta^{-1}\theta_1 + \cdots + \theta^{-1}\theta_{n-1}$ is an isomorphism, which is a contradiction to the induction assumption. $\square$

By the lemma, we find that for some s, θ_s must be an isomorphism; we may assume that $s = 1$. In this case, $V_1' = \mathrm{Im}(p_1' i_1) \oplus \mathrm{Ker}(p_1 i_1')$, so since V_1' is indecomposable, we get that $f := p_1' i_1 : V_1 \to V_1'$ and $g := p_1 i_1' : V_1' \to V_1$ are isomorphisms.

Let $B = \bigoplus_{j>1} V_j$, $B' = \bigoplus_{j>1} V_j'$; then we have $V = V_1 \oplus B = V_1' \oplus B'$. Consider the map $h : B \to B'$ defined as a composition of the natural maps $B \to V \to B'$ attached to these decompositions. We claim that h is an isomorphism. To show this, it suffices to show that $\mathrm{Ker}\, h = 0$ (as h is a map between spaces of the same dimension). Assume that $v \in \mathrm{Ker} h \subset B$. Then $v \in V_1'$. On the other hand, the projection of v to V_1 is zero, so $gv = 0$. Since g is an isomorphism, we get $v = 0$, as desired.

Now by the induction assumption, $m = n$, and $V_j \cong V_{\sigma(j)}'$ for some permutation σ of $2, \ldots, n$. The theorem is proved. $\square$

Problem 3.8.3. The above proof of Lemma 3.8.2 uses the condition that k is an algebraically closed field. Prove Lemma 3.8.2 (and hence the Krull-Schmidt theorem) without this condition.

Problem 3.8.4. (i) Let V, W be finite dimensional representations of an algebra A over a (not necessarily algebraically closed) field K. Let L be a field extension of K. Suppose that $V \otimes_K L$ is isomorphic to $W \otimes_K L$ as a module over the L-algebra $A \otimes_K L$. Show that V and W are isomorphic as A-modules.

Hint: Reduce to the case of finitely generated, then finite extension, of some degree n. Then regard $V \otimes_K L$ and $W \otimes_K L$ as